

LAB TASK WEEK 5

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Course: Object Oriented Programming Teacher:

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Task 1:

A BANK WANTS A SYSTEM THAT:

- STORES CUSTOMER INFORMATION
- MANAGES BANK ACCOUNT OPERATIONS
- ALLOWS DEPOSIT, WITHDRAWAL, AND BALANCE CHECKING

WE WILL USE TWO CLASSES:

1. CUSTOMER
2. BANK ACCOUNT

```
public static void main(String[] args) {  
  
    Customer c1 = new Customer();  
    c1.name = "Muhammad Salman";  
    c1.customerId = 72373;  
  
    BankAccount acc1 = new BankAccount();  
    acc1.accountNumber = 342533;  
    acc1.balance = 100000; acc1.customer  
    = c1;  
  
    acc1.displayAccount();  
    acc1.deposit(20000);  
    acc1.withdraw(30000);  
    acc1.displayAccount();  
}
```

MAIN CLASS

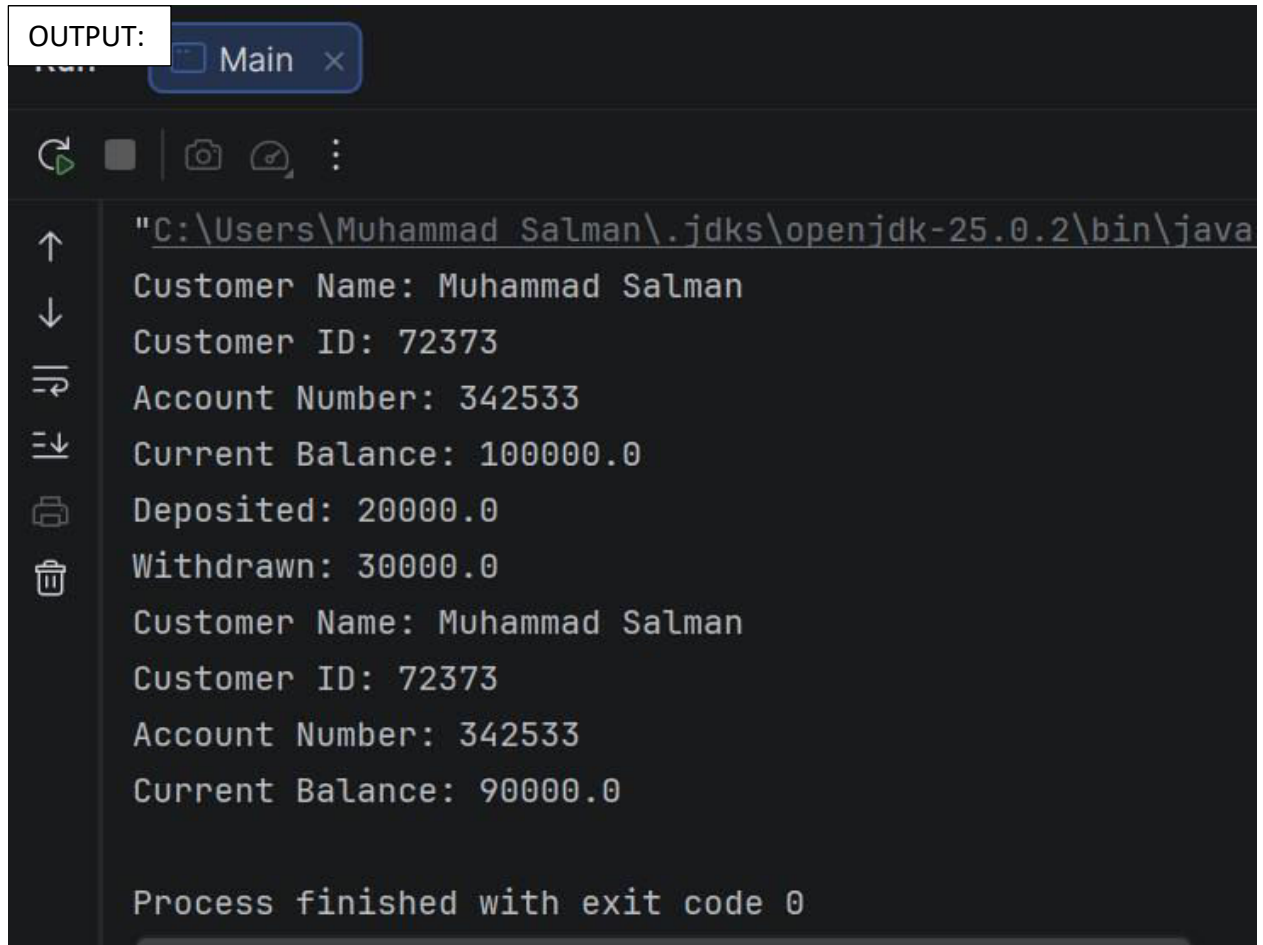
```
public class Customer {  
    String name;  
    int customerId;  
  
    void displayCustomer() {  
        System.out.println("Customer Name:" + name);  
        System.out.println("CustomerID:" + customerId);  
    }  
}
```

CUSTOMERCLASS

```
public class BankAccount {  
    int accountNumber;  
    double balance;  
    Customer customer; // Association (has-a relationship)  
  
    void deposit(double amount) {  
        balance = balance + amount;  
        System.out.println("Deposited:" + amount);  
    }  
  
    void withdraw(double amount) { if  
        (amount <= balance) {  
            balance = balance - amount;  
            System.out.println("Withdrawn:" + amount);  
        } else {  
            System.out.println("Insufficient Balance!");  
        }  
    }  
  
    void displayAccount()  
    { customer.displayCustomer();  
      System.out.println("Account Number:" + accountNumber);  
      System.out.println("Current  
      Balance:" + balance);  
    }  
}
```

BANKACCOUNTCLASS

OUTPUT:



The screenshot shows a Java IDE's output window titled "Main". The output text is as follows:

```
"C:\Users\Muhammad Salman\.jdk\openjdk-25.0.2\bin\java
Customer Name: Muhammad Salman
Customer ID: 72373
Account Number: 342533
Current Balance: 100000.0
Deposited: 20000.0
Withdrawn: 30000.0
Customer Name: Muhammad Salman
Customer ID: 72373
Account Number: 342533
Current Balance: 90000.0

Process finished with exit code 0
```

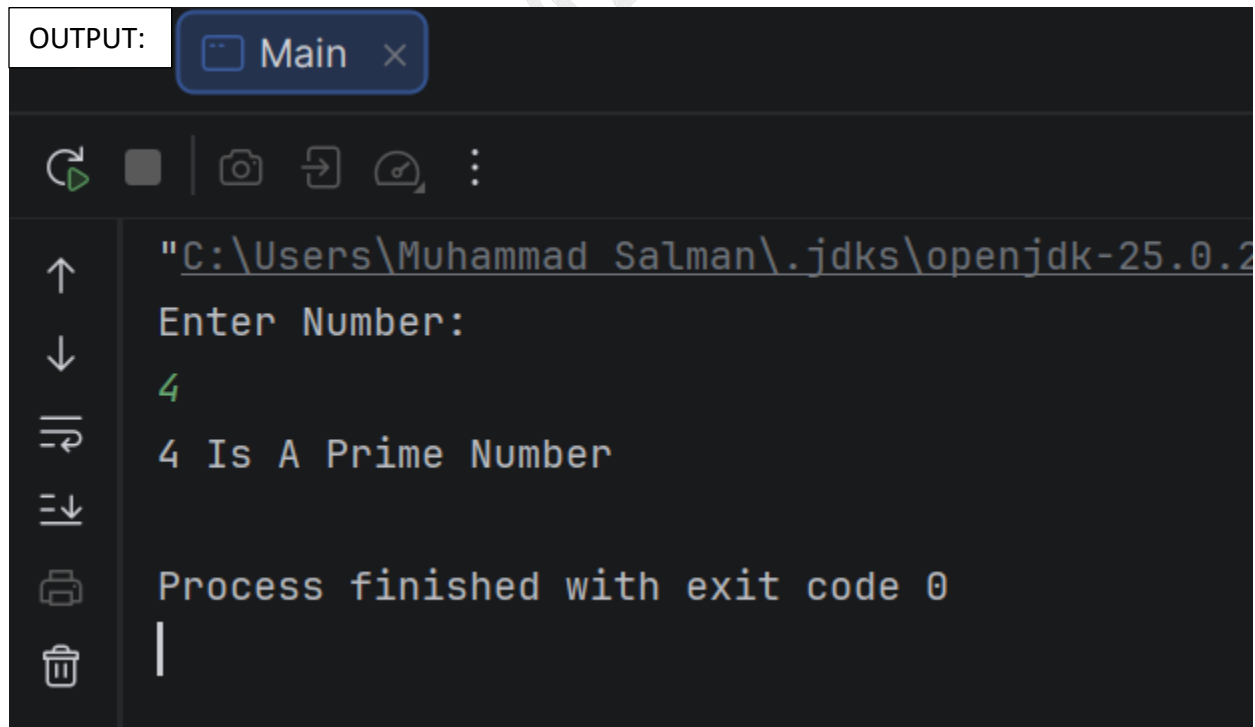
A large, light gray watermark reading "MUHAMMAD" is visible diagonally across the lower half of the image.

Task2:**PRIME NUMBER PROGRAM IN JAVA**

```
import java.util.Scanner;
public class Main {
    public static void main() {
        Scanner s = new
        Scanner(System.in); System.out.println(
        "Enter Number:"); int n;
        int
        count = 0; n = s.nextIn
        t();
        for (int i = 1; i < n; i++) {
            if (n % i == 0)
                count++;
        }
        if (count == 2)
            System.out.println(n + "Is A Prime Number"); else
            System.out.println(n + "Is Not A Prime Number");
        }
    }
}
```

CODE:

OUTPUT:



```
"C:\Users\Muhammad Salman\.jdk\openjdk-25.0.2
Enter Number:
4
4 Is A Prime Number
Process finished with exit code 0
|
```

Task3:**FIND LENGTH OF ARRAY**

<pre>public class Main { public static void main(String[] args) { int[] a = {1, 2, 8, 6, 9, 10}; int length; length = a.length; System.out.println("Length Of Array Is: " + length); } }</pre>	CODE:
--	-------

OUTPUT:

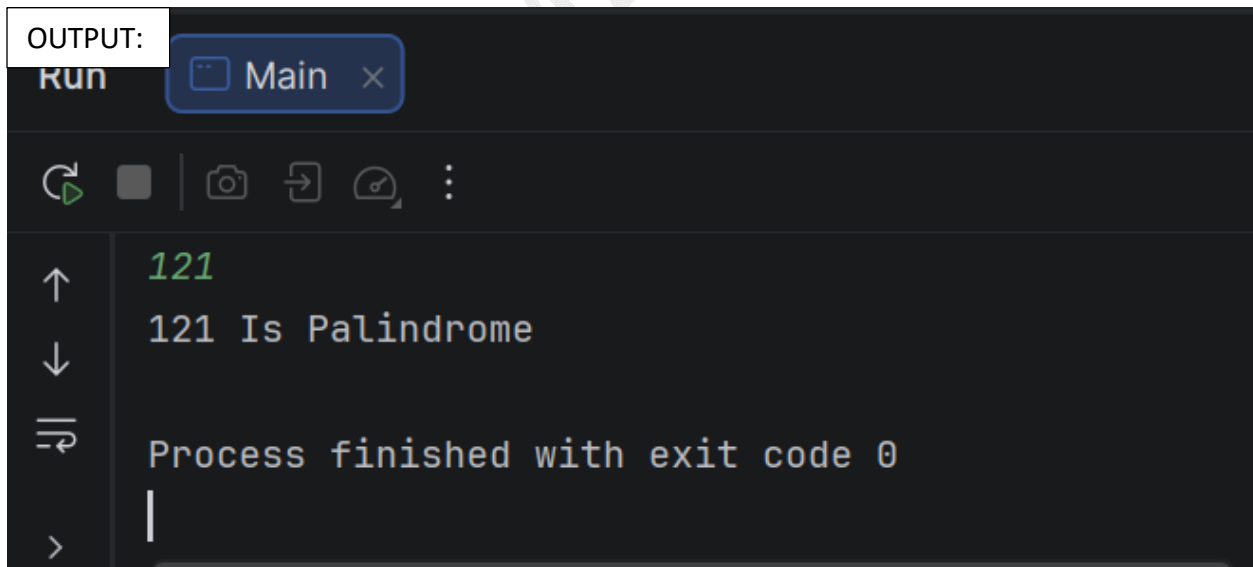


```
"C:\Users\Muhammad Salman\.jdk\openjdk-25.0.2\bin\  
Length Of Array Is:6  
  
Process finished with exit code 0
```

TASK4:**FIND THE NUMBER IS PALINDROME OR NOT**

<pre>import java.util.Scanner; public class Main { public static void main(String[] args) { Scanner s = new Scanner(System.in); int num, temp, r = 0; System.out.println("Enter Number:"); num = s.nextInt(); temp = num; while (num > 0) { r = r * 10 + (num % 10); num /= 10; } if (temp == r) System.out.println(temp + "Is Palindrome"); else System.out.println(temp + "Is Not Palindrome"); } }</pre>	CODE:
---	-------

OUTPUT:



The screenshot shows a Java IDE's output window. At the top, there's a tab labeled 'Main' with a close button. Below the tab, there's a toolbar with icons for running, stopping, and other actions. The output area shows the following text:

```
121
121 Is Palindrome
Process finished with exit code 0
```

TASK5:

FINDFACTORIALOFANUMBERTHROUGHFUNCTION

```
import java.util.Scanner;
public class Main {
    static void fact() {
        Scanner s = new Scanner(System.in);
        int a;
        System.out.println("Enter Number For Factorial:");
        a = s.nextInt();
        int fact = 1;
        for (int i = 1; i <= a; i++)
            fact *= i;
        System.out.println("Factorial Of " + a + " Is " + fact);
    }
    public static void main(String[] args) {
        fact();
    }
}
```

CODE:

OUTPUT:

Main



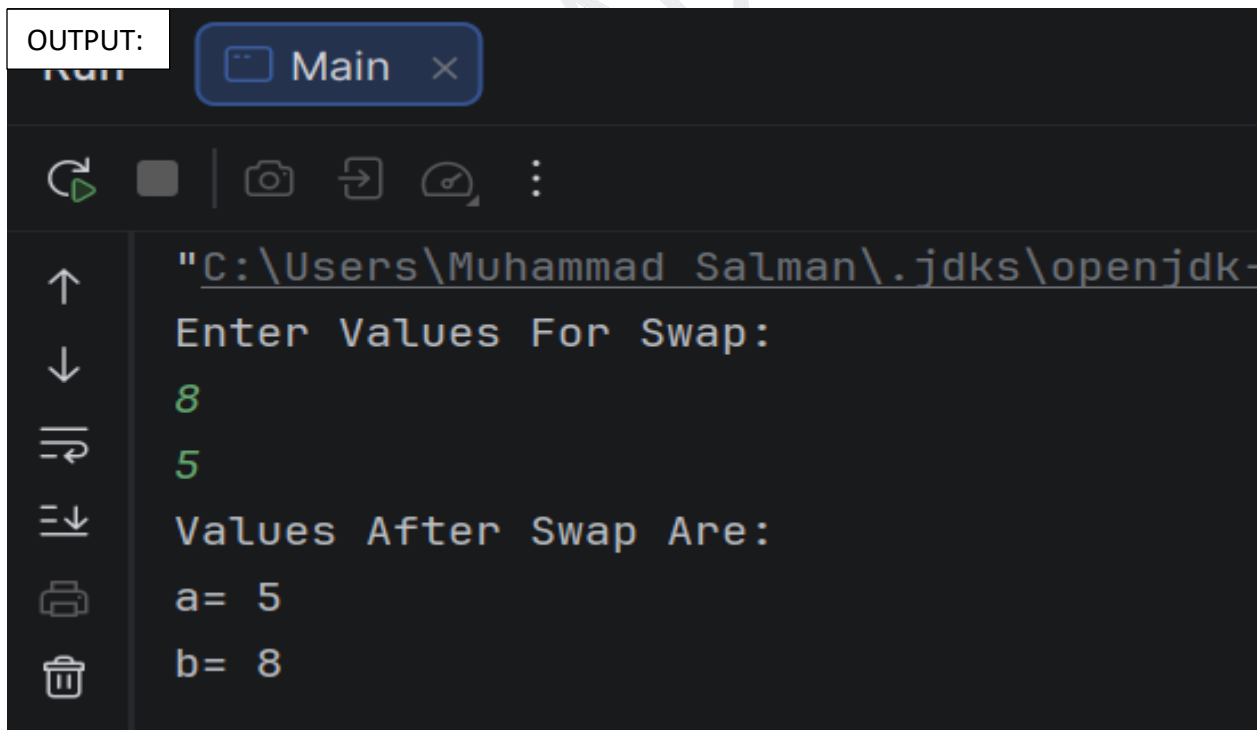
```
"C:\Users\Muhammad Salman\.jdk\openjdk-25.0.2\b  
Enter Number For Factorial:  
  
5  
Factorial Of 5 Is = 120  
  
Process finished with exit code 0
```

TASK 6:**SWAPPING NUMBERS USING TWO VARIABLES**

```
import java.util.Scanner;
public class Main {
    public static void main(String[] args) {
        Scanner s = new Scanner(System.in);
        int a, b;
        System.out.println("Enter Values For Swap:");
        a = s.nextInt();
        b = s.nextInt();
        a = a + b;
        b = a - b;
        a = a - b;
        System.out.println("Values After Swap Are:");
        System.out.println("a=" + a);
        System.out.println("b=" + b);
    }
}
```

CODE:

OUTPUT:



```
"C:\Users\Muhammad Salman\.jdk\openjdk-
Enter Values For Swap:
8
5
Values After Swap Are:
a= 5
b= 8
```


Task7:**SWAP VALUES USING THIRD VARIABLE**

<pre>import java.util.Scanner; public class Main { public static void main(String[] args) { Scanner s = new Scanner(System.in); int a, b, temp; System.out.println("Enter Values For Swap:"); a = s.nextInt(); b = s.nextInt(); temp = a; a = b; b = temp; System.out.println("Values After Swap Are:"); System.out.println("a=" + a); System.out.println("b=" + b); } }</pre>	CODE:
---	-------

OUTPUT:



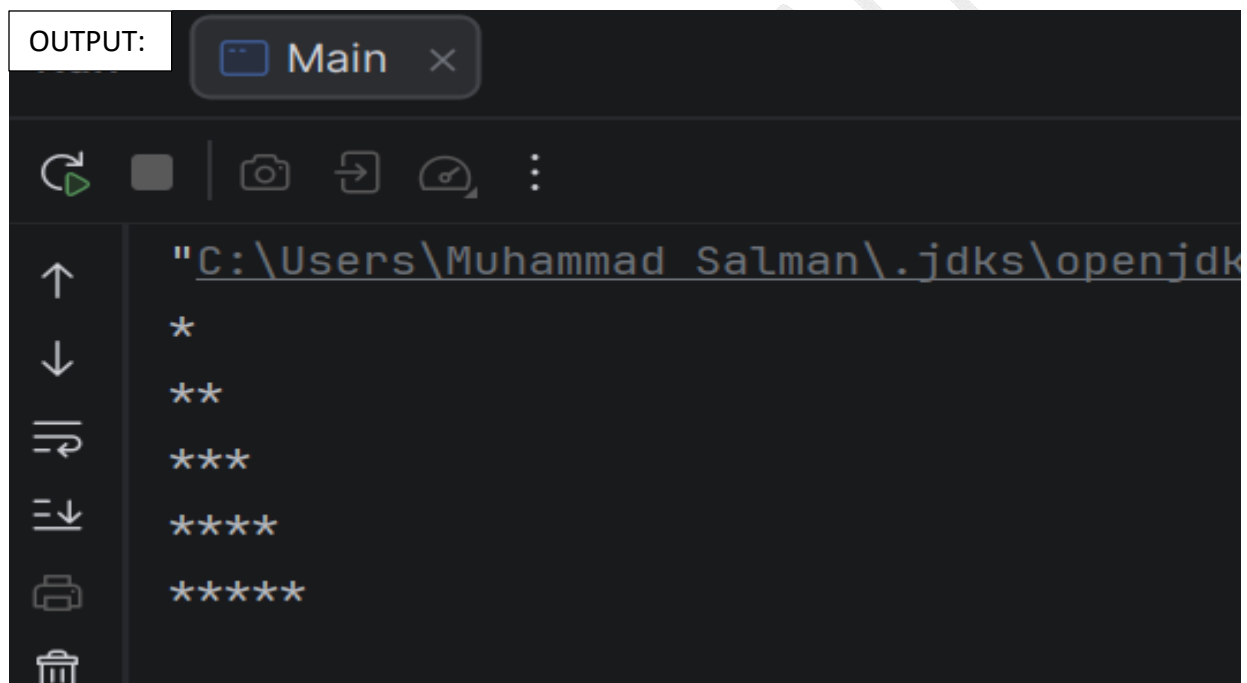
```
"C:\Users\Muhammad Salman\.jdk\openjdk-25.0.1
Enter Values For Swap:
8
5
Values After Swap Are:
a= 5
b= 8
```

Task8:**RIGHT ANGLED TRIANGLE PATTERN USING NESTED LOOP**

```
public class Main {  
    public static void main(String[] args) {  
        for (int i = 1; i <= 5; i++) {  
            for (int j = 1; j <= i; j++) {  
                System.out.print("*");  
            }  
            System.out.println();  
        }  
    }  
}
```

CODE:

OUTPUT:



```
"C:\Users\Muhammad Salman\.jdk\openjdk  
*  
**  
***  
****  
*****
```

MUHAMMAD SALMAN